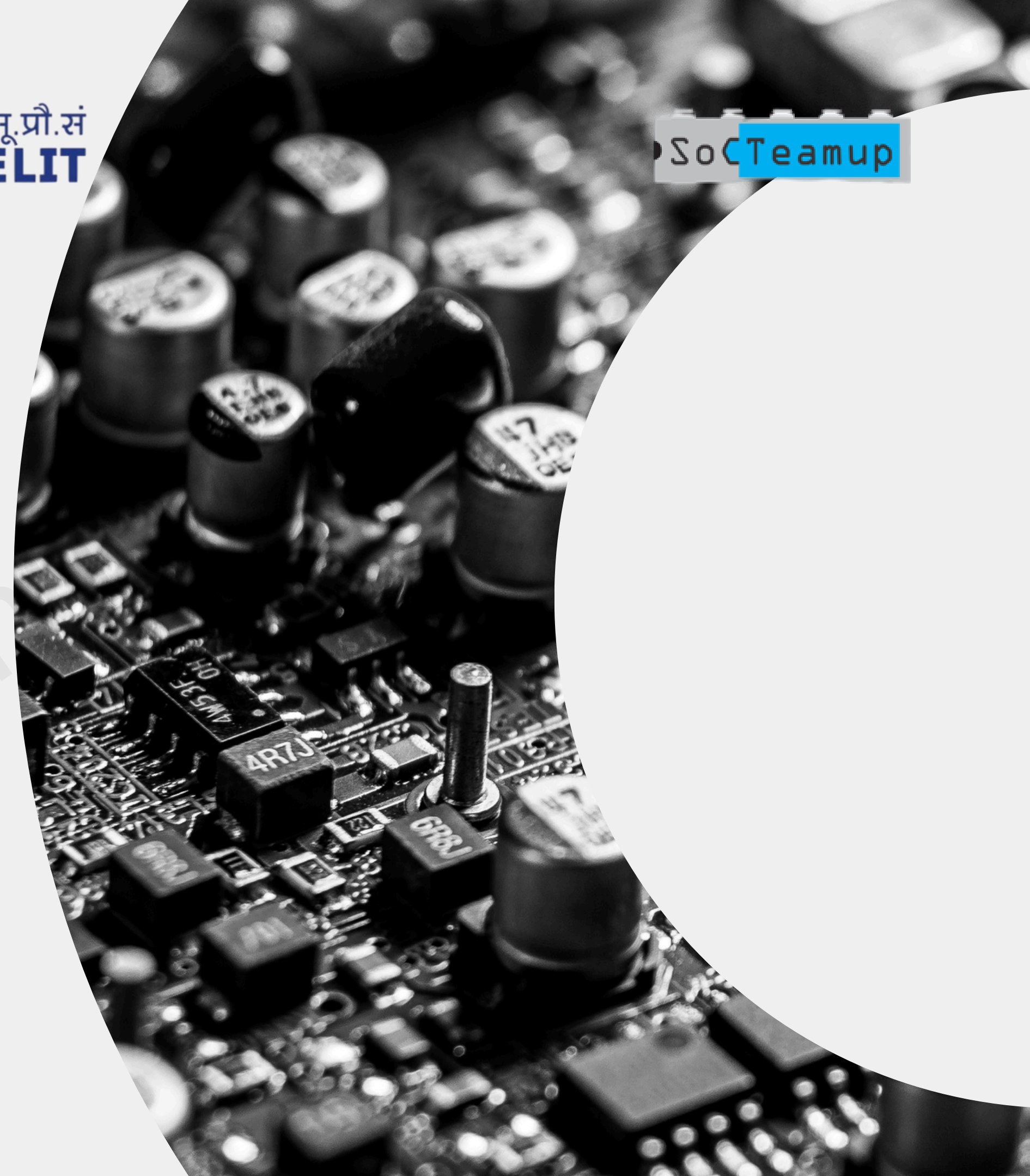


IP Integration At Subsystem Level





Outline

Why IP Integration Matters?

Types of Interfacing: Point-to-Point vs Bus-Based

Common Bus Protocols (AXI, APB, AHB)

Signal Naming and Port Mapping

Clock & Reset Domain Handling

Hierarchical Integration Example

Why IP Integration Matters?

- IPs rarely operate standalone in real designs



**INTEGRATE MULTIPLE IPS
INTO A SUBSYSTEM (E.G.,
CPU + UART + TIMER)**



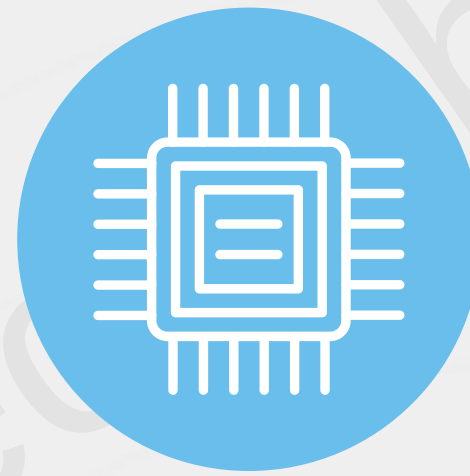
**INTERFACE MISMATCHES,
BUS PROTOCOL ERRORS
CAUSE BUGS**

Why IP Integration Matters?

- IPs rarely operate standalone in real designs



**CLEAN INTEGRATION =
FASTER BRING-UP + EASIER
DEBUG**



**GOAL:
MAKE MULTIPLE IPS WORK
TOGETHER AS ONE
COHESIVE UNIT**

➔ Interfacing Types

Point-to-Point vs. Bus-Based

Point-to-Point:	Bus-Based:
• Direct signal-level connection (e.g., UART_TX → UART_RX) • Simple but doesn't scale well	• Shared data/address/control lines (e.g., AXI4, APB) • Supports scalable SoC integration

Common On-Chip Bus Protocols

AXI4 (Advanced eXtensible Interface)	High-performance, supports burst transfers, out-of-order completion
AHB (Advanced High-performance Bus)	Pipelined transfers, used in mid-range systems
APB (Advanced Peripheral Bus)	Simple, low-power interface for slow peripherals

Signal Naming and Port Mapping

Best Practices:

- Use consistent prefixes: e.g., Uart_tx, gpio_in, axi_awvalid
- Map ports clearly using top-level wrapper modules
- Avoid name collisions across IPs

Example:

- cpu0_axi_master.awvalid →
uart0_axi_slave.awvalid

Clock & Reset **Domain** Handling

- Subsystems may run on different clocks/resets
- Need CDC (Clock Domain Crossing) logic
- Resets should be synchronized or isolated
- **Tips:**
 - Use synchronizers for control signals
 - Use asynchronous FIFOs for data paths
 - Separate clean resets for each IP domain

Hierarchical Integration Example

Subsystem Block Example:	Integration Files:
• CPU (AXI Master) • UART (APB Slave) • Timer (APB Slave) • Interconnect logic (AXI → APB bridge)	• Top-level RTL wrapper • Bus decoder or fabric • IP parameter overrides

Thank You

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